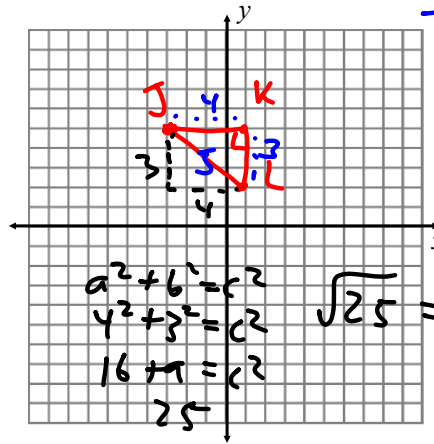


Students will calculate the perimeter & area of polygons.

Do Now

- 1) Find the perimeter and area of $\triangle JKL$ with coordinates $J(-3,5)$, $K(1,5)$, and $L(1,2)$.



P:

$$P = 4 + 3 + 5$$

$$P = 12 \text{ units}$$

A:

$$A = \frac{1}{2} b \cdot h$$

$$= \frac{1}{2} (3)(4)$$

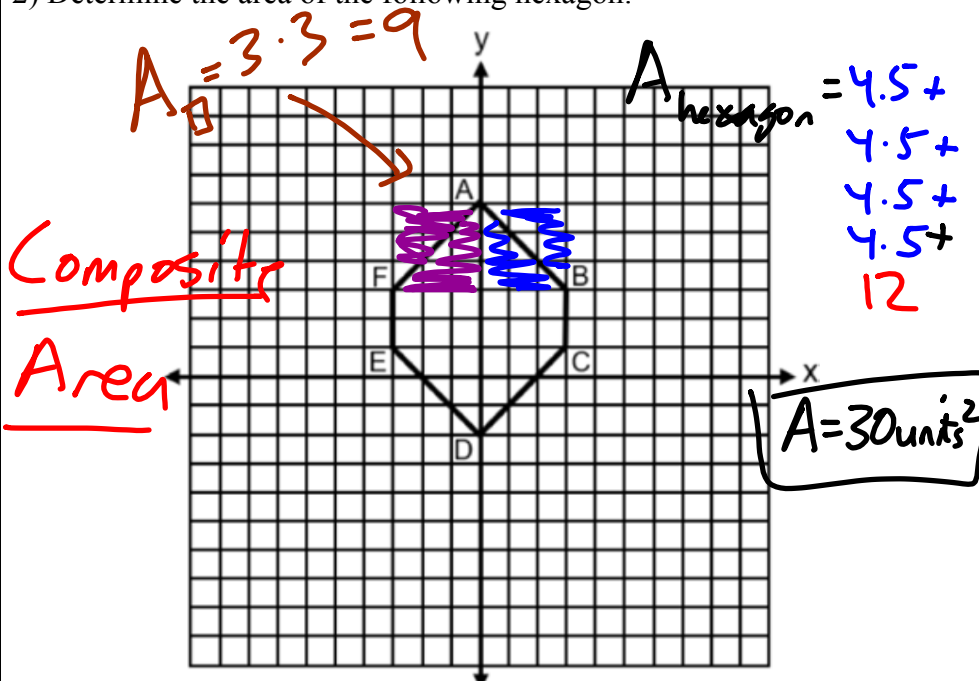
$$= 6 \text{ units}^2$$

$$\begin{aligned} a^2 + b^2 &= c^2 \\ 4^2 + 3^2 &= c^2 \\ 16 + 9 &= c^2 \\ 25 & \end{aligned}$$

$$\sqrt{25} = 5$$

Do Next

- 2) Determine the area of the following hexagon:



$$A_{\square} = 3 \cdot 3 = 9$$

$$\begin{aligned} A_{\text{hexagon}} &= 4.5 + \\ &4.5 + \\ &4.5 + \\ &4.5 + \\ &12 \end{aligned}$$

Composite
Area

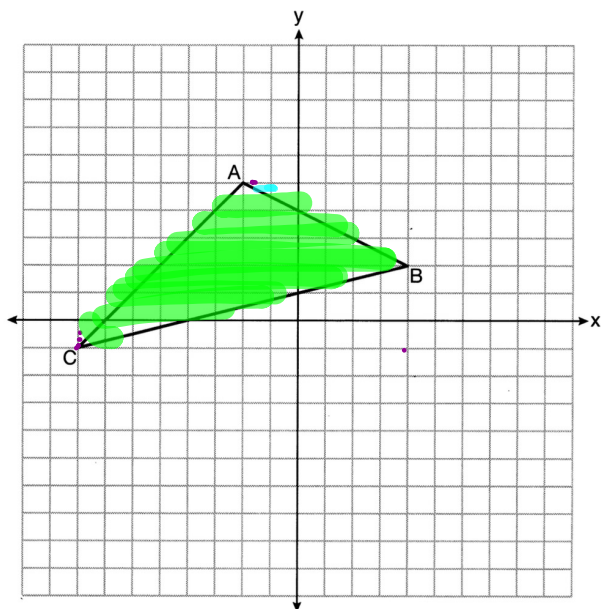
$$A = 30 \text{ units}^2$$

Done early? Determine the hexagon's perimeter.

Ex 1: **Rough Draft**

Area of Polygons

Triangle ABC with coordinates $A(-2,5)$, $B(4,2)$, and $C(-8,-1)$ is graphed on the set of axes below.



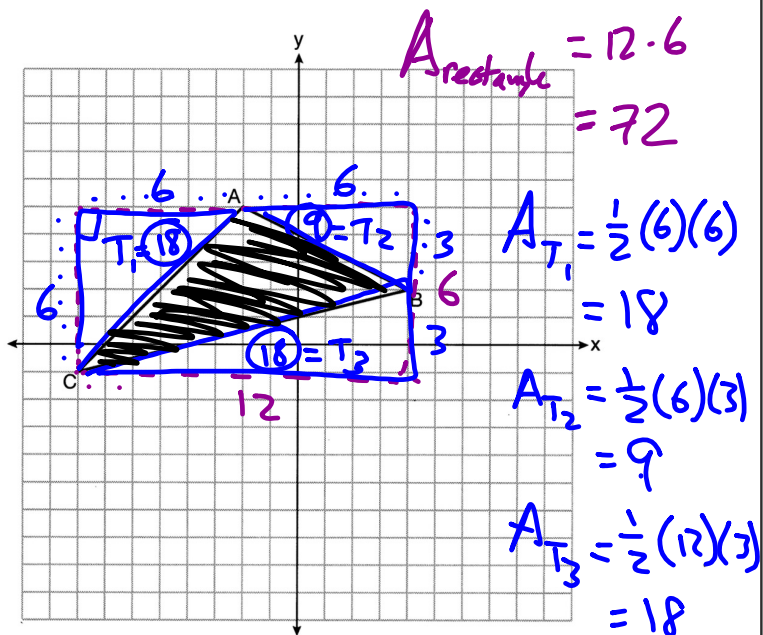
Determine and state the area of $\triangle ABC$.

Use this space for any calculations.

Ex 1: Final Draft

Area by Encasement

Triangle ABC with coordinates $A(-2,5)$, $B(4,2)$, and $C(-8,-1)$ is graphed on the set of axes below.



Determine and state the area of $\triangle ABC$.

Use this space for any notes.

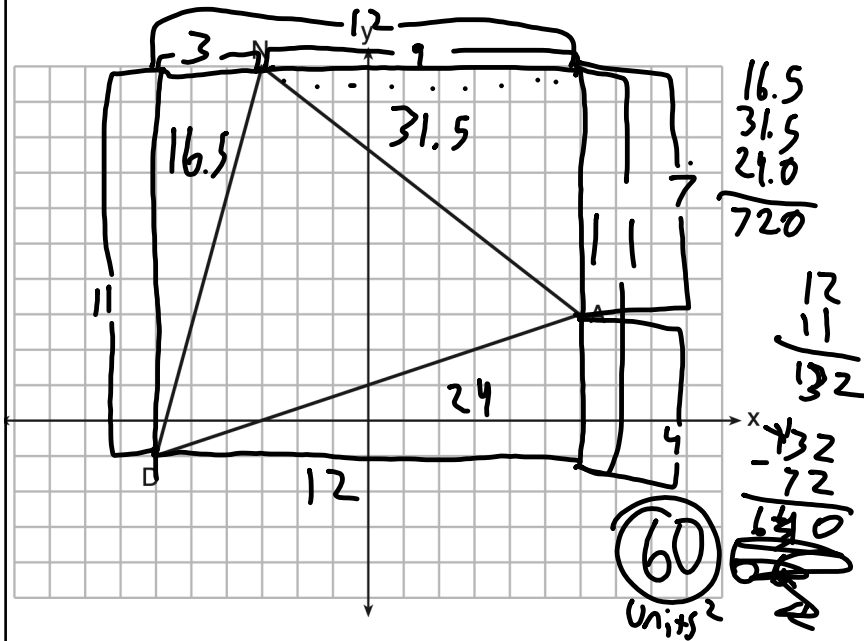
$$A_{\triangle ABC} = A_{\text{rectangle}} - A_{T_1} - A_{T_2} - A_{T_3}$$

$$= 72 - 18 - 9 - 18$$

$$A_{\triangle ABC} = 27 \text{ units}^2$$

Ex 2:

Triangle DAN is graphed on the set of axes below. The vertices of $\triangle DAN$ have coordinates $D(-6, -1)$, $A(6, 3)$, and $N(-3, 10)$.

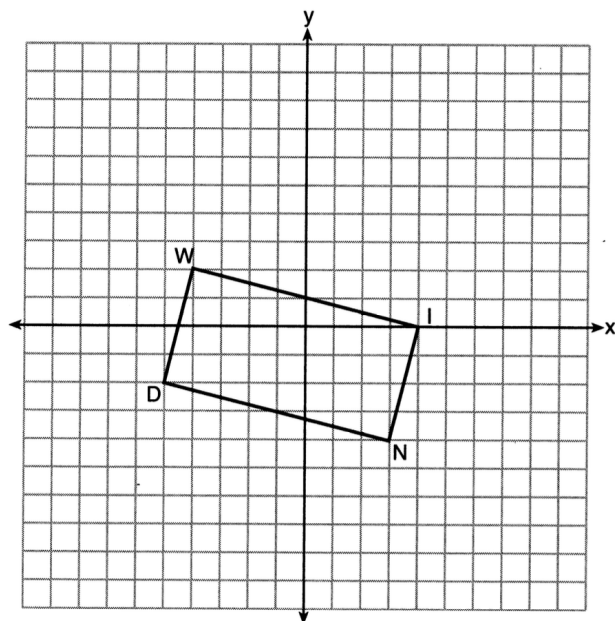


What is the area of $\triangle DAN$?

Use this space for any calculations.

Ex 3:

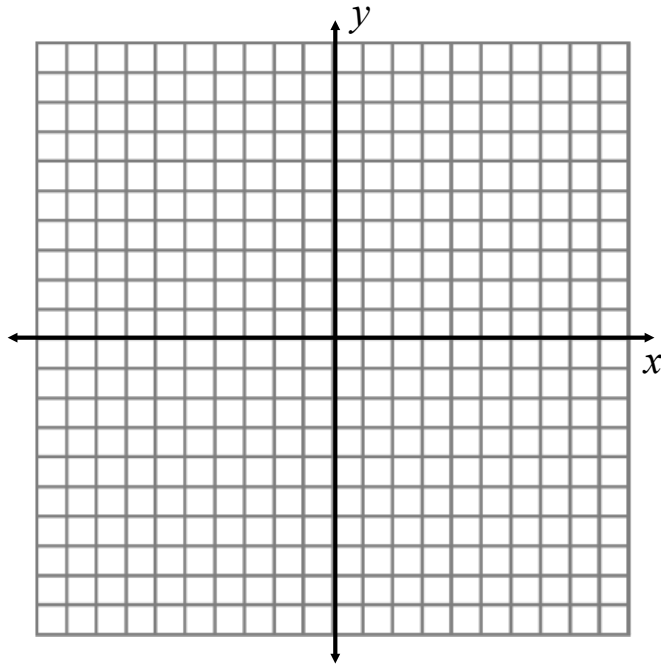
On the set of axes below, rectangle $WIND$ has vertices with coordinates $W(-4,2)$, $I(4,0)$, $N(3,-4)$, and $D(-5,-2)$.



What is the area of rectangle $WIND$?

Use this space for any calculations.

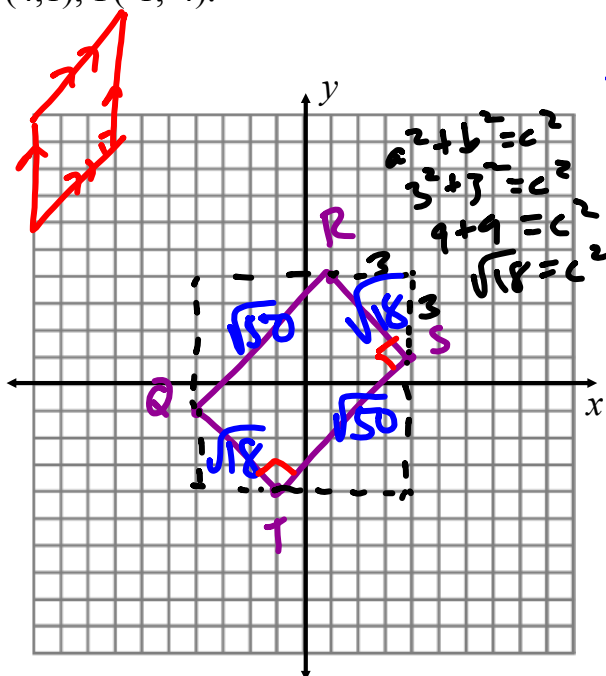
Ex 4: The points $D(9,-4)$, $E(6,8)$, $F(-3,6)$, and $G(0,-6)$ form quadrilateral $DEFG$. Determine the area of $DEFG$.



Use this space for any calculations.

Finish early? Find the perimeter of the shapes shown in Examples 1 - 4.

5) Find the perimeter and area of $QRST$ with coordinates $Q(-4,-1)$, $R(1,4)$, $S(4,1)$, $T(-1,-4)$.



P:

$$P = \sqrt{18} + \sqrt{50} + \sqrt{18} + \sqrt{50}$$

A:

$$A = \sqrt{18} \sqrt{50}$$

$$= 30$$